

JANANI ASOKAN

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SUMMARY:

Passionate software engineer with hands-on experience in Software Testing, AI/ML. Proficient in Python and C++ with a focus on building scalable and Automation Test Scripts. Applying my technical skills to drive innovation in next-generation computing technologies

EXPERIENCE:

Software and System Test Intern, Signify , Bengaluru, India July 2025 – May 2026

- Built a scalable cloud-native API automation Test Script using Python and Pytest to validate RESTful APIs across dev, staging, and production, improving test coverage, reliability, and release quality.
- Validating data flow between MQTT brokers, cloud APIs, and mobile clients to ensure system reliability and synchronization.
- Implemented robust authentication and authorization handling (dynamic tokens, session management) and resilience mechanisms such as retries and response-time validation.
- Executed firmware test cases using MQTT protocol for real-time device communication ensuring reliable message validation

EDUCATION:

MTech Computer Science and Engineering | CGPA: 8.4 | Vellore Institute of Technology, Vellore July 2021 – July 2026

SKILLS:

Languages: Python, C, C++

AI Tech: Machine Learning, Deep Learning, NLP, LLM, RAG, Generative AI, Agentic AI

Web Development: HTML, CSS, JavaScript, MERN

Frameworks: React.js, TensorFlow, scikit-learn, NumPy, Pandas, OpenCV, Matplotlib

Data Visualization: Exploratory Data Analysis (EDA), Power BI, Object Oriented Analysis and Design

Dev Tools: Docker , Kubernetes, CI/CD, DevOps, RESTful API, Wireshark ,Packet Sender

Software: GitHub, Linux/Unix, VS Code, Google Colab, Postman API, Pytest, Firmware, MQTT

Computer Science Fundamentals: Data Structures and Algorithms, Operating Systems, Computer Networks, AWS

Soft Skills: Clear technical communication, efficient debugging, consistent contributions to team goals

PROJECTS:

Adaptive Traffic Signal Control System for Emergency Vehicle Prioritization using IoT | AWS IoT, YOLO, ML

- Engineered an IoT-based traffic control system using Q-Learning to prioritize emergency vehicles, tested on 90-second video feeds.
- Facilitated automatic signal switching at 4-way intersections, cutting manual intervention and improving emergency response time by 30%.

Memory-Optimisation-in-Android-Phones-using-AI-ML. | Hugging Face, OneDal, Intel OneApi

- Developed MemoryMate, an Android application that optimizes cache memory usage to enhance phone performance
- Implemented a Random Forest predictive model to analyze cache parameters, memory access patterns, and predict miss rates.
- Provided data-driven insights and performance graphs, enabling informed memory optimization and enhancing overall device efficiency by up to 30%

Real-Time Handwritten Alphabet Recognition System and Object Detection | Deep Learning, Open CV

- Constructed a real-time handwritten recognition system with OpenCV and EMNIST, achieving 85% accuracy and identifying 20+ object classes.

IoT-Based Response Drone for Search and Rescue | React.js, Arduino,Embeeded systems,ML,Drone

- Developed an IoT-enabled drone system using Arduino for real-time disaster search and rescue operations
- Integrated an AI-driven image processing module to identify stranded individuals in disaster-affected areas with 95% accuracy.
- Conducted extensive field tests, simulations in flood conditions, tested the drone’s performance 18,000 km away, demonstrating global operability. Enabled remote operation and deployment anywhere in the world

WebDigestAI: Efficient Content Digesting with AI-driven Insights | NLP, Python, Langchain, Langgraph

- Engineered an AI-driven article summarization tool that extracts content from diverse web sources, reducing time by 65%.
- Implemented real-time data scraping from 1000+ URLs using BeautifulSoup4 and WebBaseLoader
- Applied LangChain and RAG Model Methodology for advanced NLP. Integrated the Gemini API for 90% accurate responses.

AWARDS AND HONORS

- **PitchWar Hackat finalist** secured 3rd place by developing an IoT-enabled drone system with AI-driven image processing - Best Innovator Award
- **ISRO Hackathon finalist**, achieving **87.2% accuracy** in 30k moon image annotation using YOLO V8 [Link].
- **GEO PREDICT HACK**– ML powered web application for predicting geographical trends using real-time data analysis [\[LINK\]](#)
- **Research on Federated Learning** for Autonomous Vehicle Detection– Developed a decentralized AI model
- CouchDatabase - CouchDb(Remote Internship) [\[LINK\]](#)
- ISTQB Certified Tester - [\[LINK\]](#)